

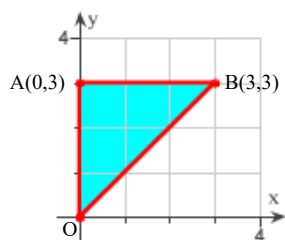
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Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find the absolute maximum and minimum of the function $f(x,y) = 2x^2 - 12x + y^2 - 6y + 10$ on the closed triangular plate bounded by the lines $x = 0$, $y = 3$, and $y = x$ in the first quadrant.

The domain R is shown below. Notice that the vertices have been labeled.



To find the absolute extrema of a continuous function $f(x,y)$ on a closed and bounded region R , first list the interior points of R where f may have local maxima and minima and evaluate f at these points. These are the critical points of f (interior points where $f_x = f_y = 0$ or points where f_x or f_y fail to exist).

First find f_x .

$$\begin{aligned} f_x &= \frac{\partial f}{\partial x} \\ &= \frac{\partial}{\partial x} (2x^2 - 12x + y^2 - 6y + 10) \\ &= 4x - 12 \end{aligned}$$

Next find f_y .

$$\begin{aligned} f_y &= \frac{\partial f}{\partial y} \\ &= \frac{\partial}{\partial y} (2x^2 - 12x + y^2 - 6y + 10) \\ &= 2y - 6 \end{aligned}$$

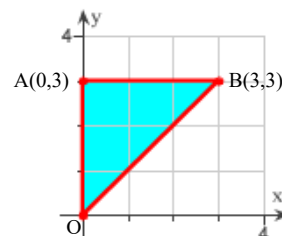
Now set both of these partial derivatives equal to zero.

$$\begin{aligned} 4x - 12 &= 0 \\ 2y - 6 &= 0 \end{aligned}$$

Solve the first equation for x and the second equation for y .

$$\begin{aligned} 4x - 12 &= 0 & 2y - 6 &= 0 \\ x &= 3 & y &= 3 \end{aligned}$$

The point $(3,3)$ is on the boundary of R , not in the interior. Therefore, it needn't be considered yet.



Now consider the boundary of R one segment at a time.

First consider the segment OA . On this segment, $x = 0$. Let $g(y) = f(0,y)$. The expression for f can be rewritten as follows.

$$\begin{aligned} f(x,y) &= 2x^2 - 12x + y^2 - 6y + 10 \\ g(y) &= f(0,y) \\ &= y^2 - 6y + 10 \end{aligned}$$

The extreme values of $g(y) = y^2 - 6y + 10$ may occur at the endpoints of the segment OA or at the interior points where $g'(y) = 0$.

Compute the values of $g(y)$ at $y = 0$ and $y = 3$, the endpoints of the segment OA .

$$g(y) = y^2 - 6y + 10$$

$$g(y) = y^2 - 6y + 10$$

$$\begin{aligned} g(0) &= (0)^2 - 6(0) + 10 \\ &= 10 \end{aligned}$$

$$\begin{aligned} g(3) &= (3)^2 - 6(3) + 10 \\ &= 1 \end{aligned}$$

Now find $g'(y)$.

$$g(y) = y^2 - 6y + 10$$

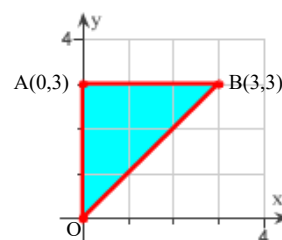
$$g'(y) = 2y - 6$$

Set $g'(y) = 0$ and solve for y .

$$2y - 6 = 0$$

$$y = 3$$

Notice that $y = 3$ corresponds to one of the endpoints of the segment OA. As computed above, $g(3) = 1$.



Next consider the segment AB. On this segment, $y = 3$. Let $h(x) = f(x, 3)$. The expression for f can be rewritten as follows.

$$f(x, y) = 2x^2 - 12x + y^2 - 6y + 10$$

$$h(x) = f(x, 3)$$

$$= 2x^2 - 12x + 1$$

The extreme values of $h(x) = 2x^2 - 12x + 1$ may occur at the endpoints of the segment AB or at the interior points where $h'(x) = 0$.

Compute the values of $h(x)$ at $x = 0$ and $x = 3$, the endpoints of the segment AB. Notice that $h(0) = g(3)$ and was computed above.

$$h(x) = 2x^2 - 12x + 1$$

$$h(x) = 2x^2 - 12x + 1$$

$$\begin{aligned} h(0) &= 2(0)^2 - 12(0) + 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} h(3) &= 2(3)^2 - 12(3) + 1 \\ &= -17 \end{aligned}$$

Now find $h'(x)$.

$$h(x) = 2x^2 - 12x + 1$$

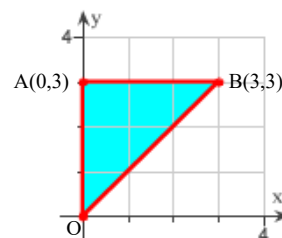
$$h'(x) = 4x - 12$$

Set $h'(x) = 0$ and solve for x .

$$4x - 12 = 0$$

$$x = 3$$

Notice that $x = 3$ corresponds to one of the endpoints of the segment AB. As computed above, $h(3) = -17$.



Now consider the segment OB. On this segment, $y = x$. Let $k(x) = f(x, x)$. Therefore, the expression for f can be rewritten as follows.

$$f(x, y) = 2x^2 - 12x + y^2 - 6y + 10$$

$$k(x) = f(x, x)$$

$$= 3x^2 - 18x + 10$$

The extreme values of $k(x) = 3x^2 - 18x + 10$ may occur at the endpoints of the segment OB or at the interior points where $k'(x) = 0$.

Both endpoints of the segment OB have already been considered.

$$k(0) = 10$$

$$k(3) = -17$$

Find $k'(x)$.

$$k(x) = 3x^2 - 18x + 10$$

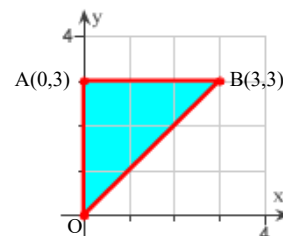
$$k'(x) = 6x - 18$$

Set $k'(x) = 0$ and solve for x .

$$6x - 18 = 0$$

$$x = 3$$

Notice that $x = 3$ corresponds to one of the endpoints of the segment OB. As computed above, $k(3) = -17$.



Therefore, the candidates for the absolute maxima and minima of f over the given domain are as follows.

$$f(0,0) = 10$$

$$f(0,3) = 1$$

$$f(3,3) = -17$$

Therefore, on the given domain, the function's absolute maximum is 10 and its absolute minimum is -17 .

The function assumes its absolute maximum at $(0,0)$ and its absolute minimum at $(3,3)$.